

Magnetone Model MP3 Fixed Bias Preamp



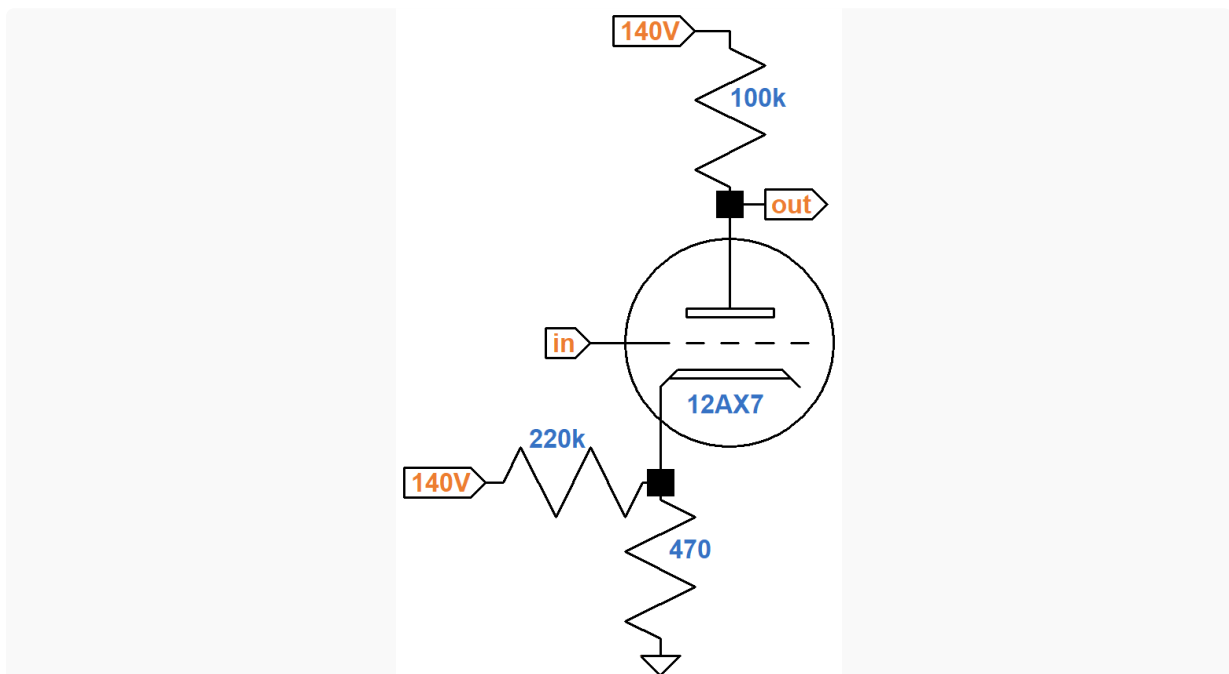
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The first stage of the Magnetone MP3 guitar amplifier adds a strange 220kΩ resistor between the 140V plate supply and the cathode.



According to Magnetone's schematic, the measured cathode voltage is 0.5V. Let's see how this matches an average 12AX7 as documented by the data sheet.

According to Ohm's Law, if we assume the cathode voltage is 0.5V then the current through the 470Ω resistor is

$$0.5V / 470\Omega = 1.06mA$$

The current through the 220kΩ resistor is

$$(140V - 0.5V) / 220k\Omega = 0.63mA$$

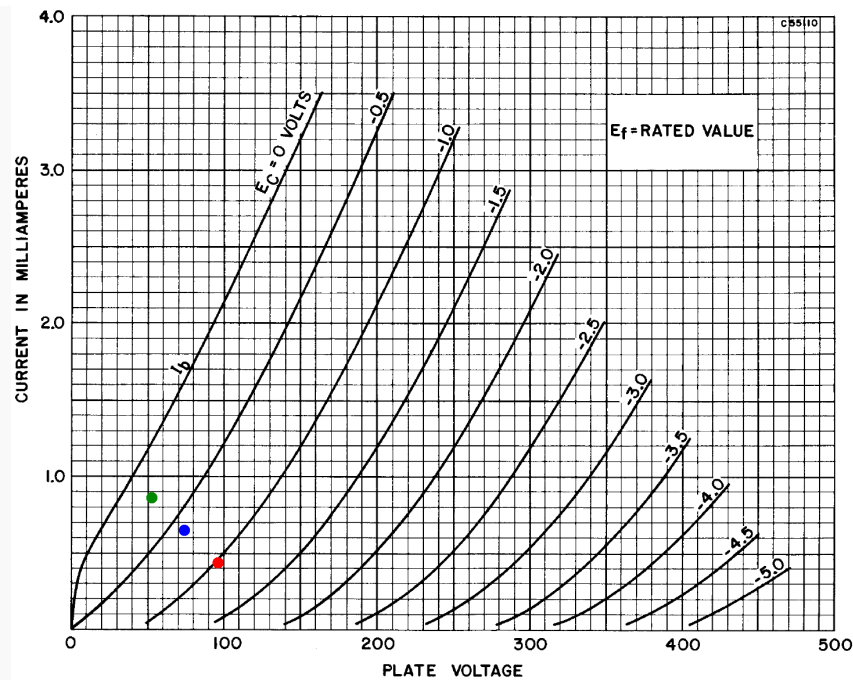
The sum of this current and the plate current passes through the 470Ω resistor, so the plate current is

$$1.06\text{mA} - 0.63\text{mA} = 0.43\text{mA}$$

The plate-to-cathode voltage is equal to the plate supply voltage, minus the voltage drop across the 100kΩ plate load resistor, minus the cathode voltage:

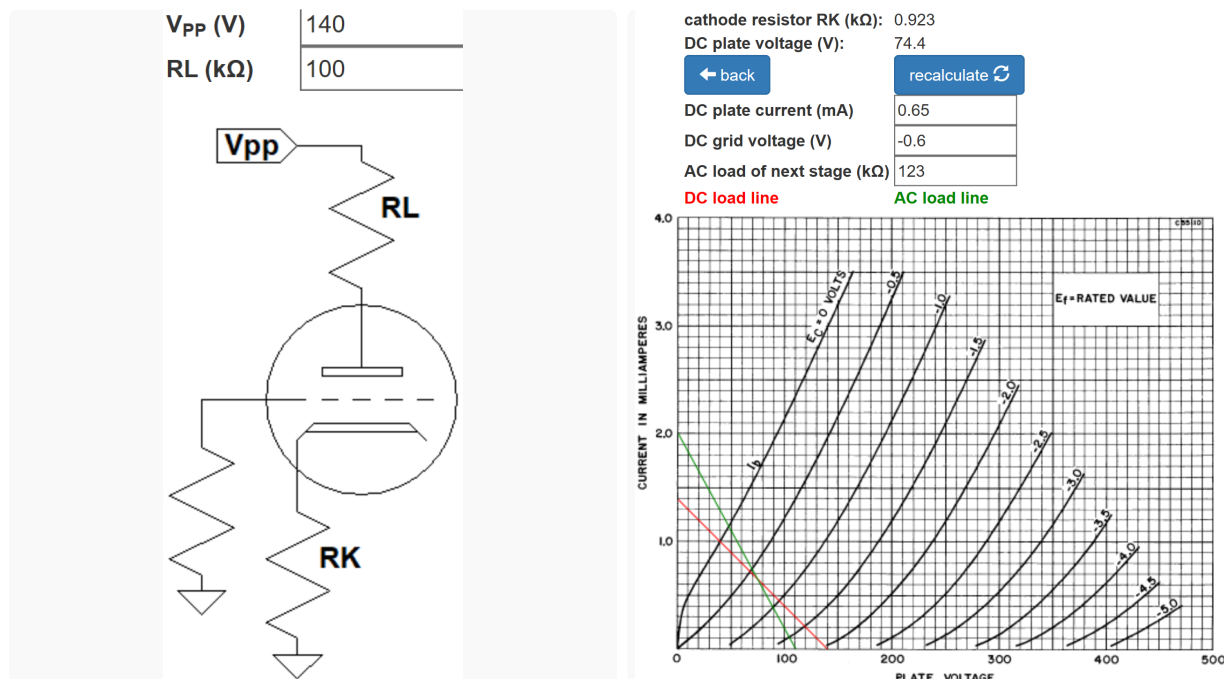
$$140\text{V} - (0.43\text{mA})(100\text{k}\Omega) - 0.5\text{V} = 96.5\text{V}$$

Here on the 12AX7 plate characteristic curves, the point for 97V and 0.43mA is marked by the red dot, which indicates the grid voltage should be slightly less than -1V for this DC operating point.



We can repeat these computations for an assumed 0.6V cathode (blue dot, 74V, 0.65mA) and an assumed 0.7V cathode (green dot, 53V, 0.86mA). The blue dot wins: it is in agreement with an imagined -0.6V curve.

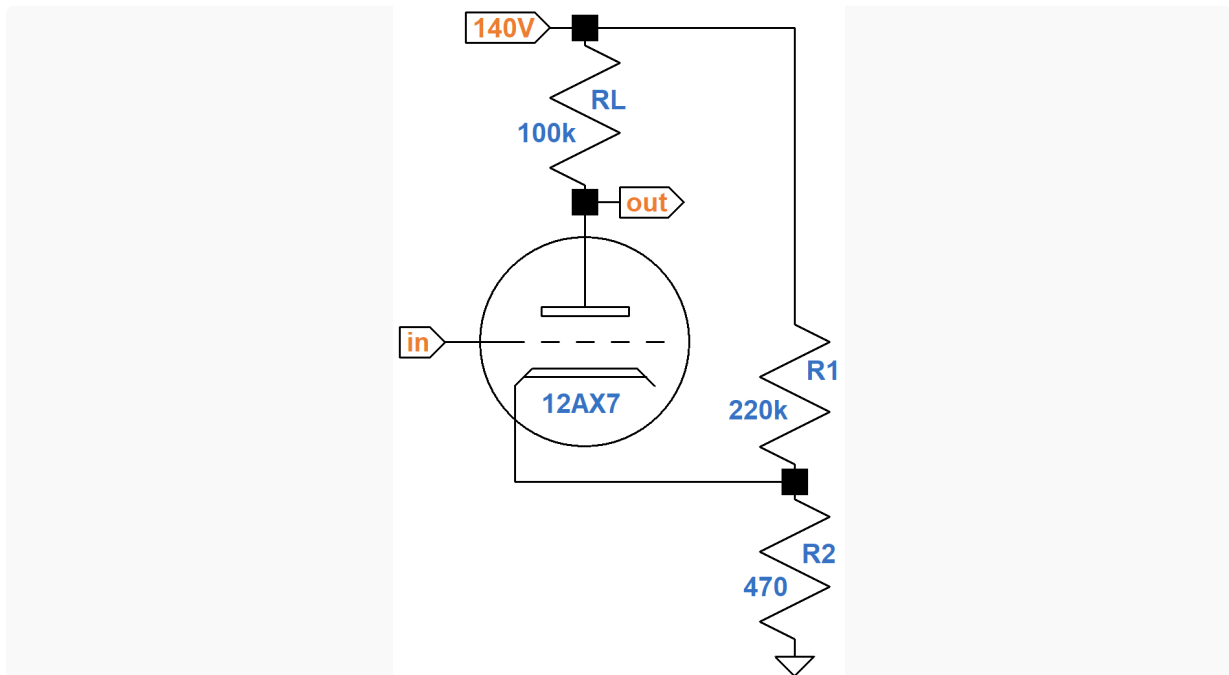
It is interesting to see that the dots form a line. In fact, the upper left endpoint of the line appears to be at a plate current of 1.4mA on the Y axis. The lower right endpoint is at a plate voltage of 140V on the X axis, so the dots are on a traditional DC load line,¹ as shown by the red line of the [12AX7 calculator](#).



The calculator shows that the same bias can be achieved with a 923Ω cathode resistor for traditional cathode bias. This is not a standard resistor value, but great precision is not the hallmark of guitar amp design. For the discussion that follows, let's assume a circuit designer would select a 1kΩ value to achieve the same DC operating point using traditional cathode bias.

Fixed-Bias

When the circuit is redrawn, Magnatone's concept becomes clear.



This is not a traditional cathode-biased preamp stage with an extra resistor. Instead, it is a pair of resistors that form a voltage divider to create *fixed bias*, as defined by the Radiotron Designer's Handbook.² Here is a step-by-step procedure to determine the value of R_2 for a fixed-bias voltage amplifier based on the plate supply voltage V_{PP} , the plate load resistor R_L , and the desired value of R_1 :

1. Draw a traditional DC load line that connects the plate supply voltage V_{PP} with V_{PP} / R_L .
2. Along the load line, pick a desired DC operating point.
3. Note the DC grid voltage V_G and the DC plate current I_P at that point.
4. Solve for R_2 using the following formula.

$$R_2 = \frac{V_K}{I_P + \frac{V_{PP} - V_K}{R_1}}$$

where the cathode voltage is

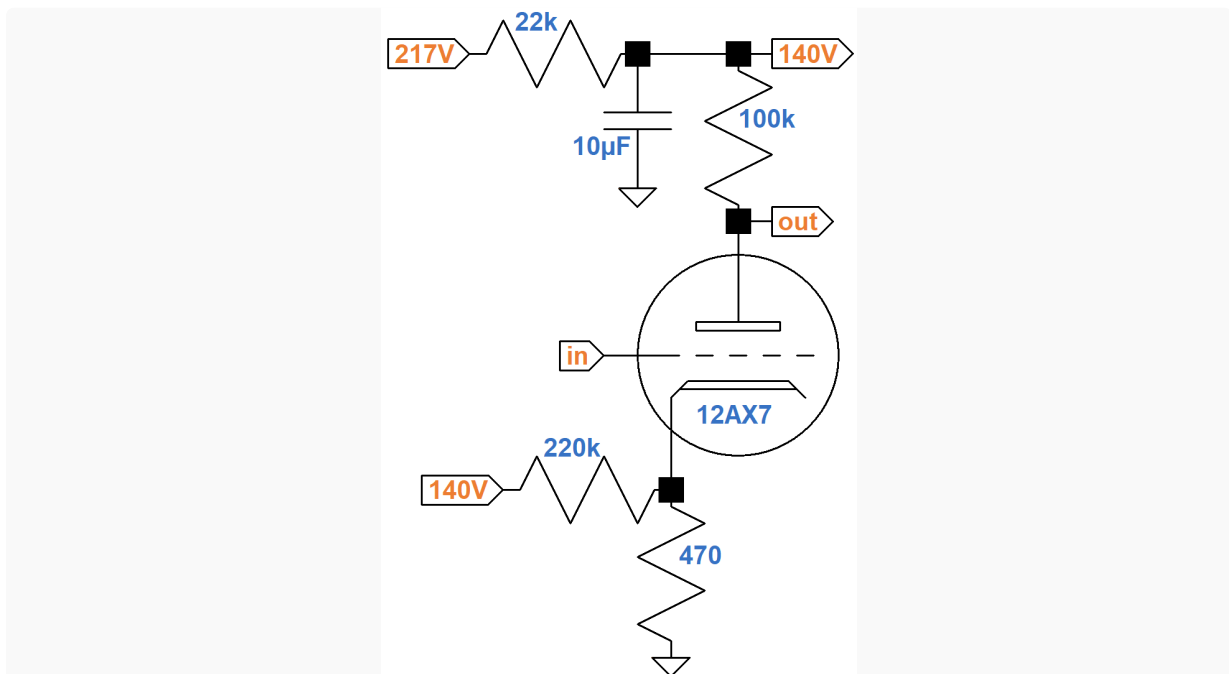
For a desired $R_1 = 220k\Omega$ and $V_K = 0.6V$, this procedure yields a result that is very close to Magnatone's value:

$$R_2 = \frac{(0.6V)}{0.65mA + \frac{140V - 0.6V}{220k\Omega}} = 467\Omega$$

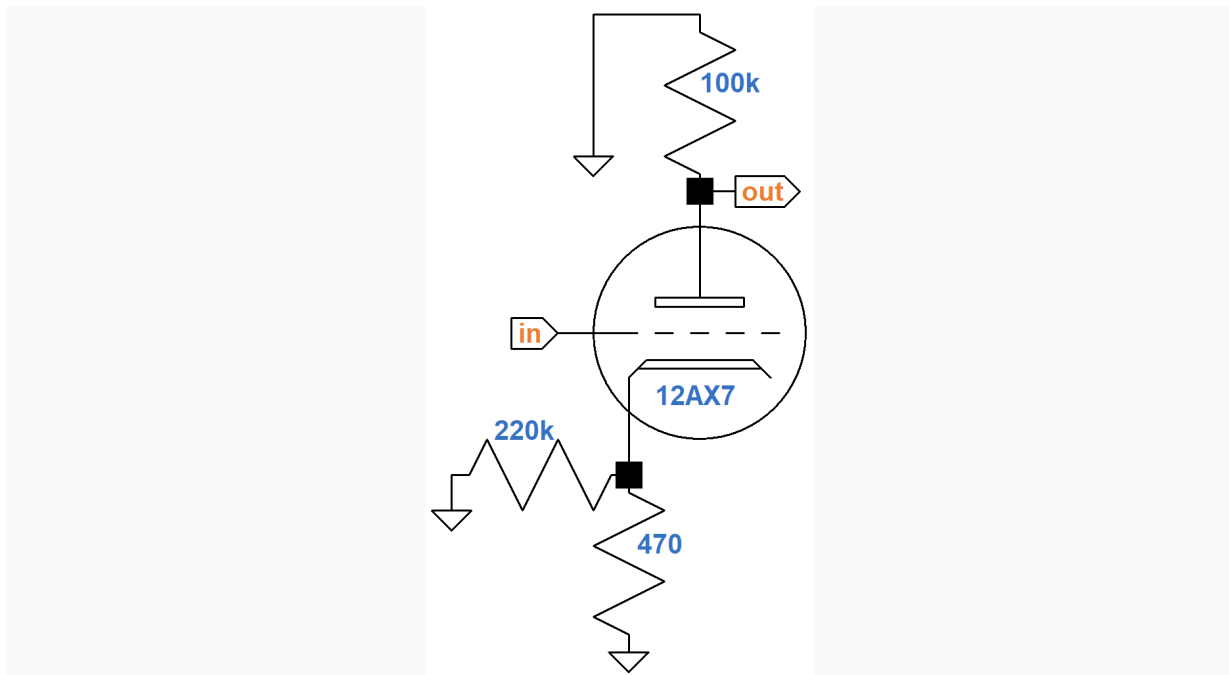
From a DC perspective, does a pair of 220kΩ and 470Ω resistors for fixed bias offer any advantage over a traditional single-resistor design? Not really, unless you have a warehouse full of them and have 1kΩ resistors on backorder. We could also use 150kΩ for R1 and 390Ω for R2. In fact, there are many combinations of values for R1 and R2 that can be used to create the same DC operating point. So perhaps one advantage might be *inventory flexibility*. Typically, R1 has a large value to limit the current that the voltage divider draws from the high-voltage power supply.

Audio Performance

Here is Magnatone's first stage again, this time including the RC ripple filter for the 140V plate supply.



For audio signals, the capacitor acts as a short circuit,³ so the equivalent AC circuit puts the 220kΩ resistor in parallel with the cathode resistor.



220k Ω is huge compared to 470 Ω , so the effective AC cathode resistor value is almost the same as without it: 469 Ω . There is no significant difference in gain when the 220k Ω resistor is included, so audio performance is the same as for traditional cathode bias **if a 470 Ω resistor is used**. To create the same DC operating point, however, we observed that cathode bias requires about **1k Ω** .

If the cathode is bypassed by a capacitor, then there is no negative feedback from cathode degeneration⁴ and audio performance is the same for either resistor value. In this case traditional cathode bias has the advantage that a smaller capacitor value can achieve full bypass. Magnatone's resistor is unbypassed, however, so gain is reduced. A 470 Ω value, however, creates less cathode degeneration than a 1k Ω value. According to the [Preamp Gain and Output Impedance calculator](#), the gain is 35.8dB when the cathode is fully bypassed. When unbypassed, gain is 33.6dB with 470 Ω and 31.6dB with 1k Ω .

From a system design⁵ perspective, greater inventory flexibility and a few less dB attenuation when unbypassed are not particularly compelling reasons to add a resistor to the circuit. It is not surprising, therefore, that Magnatone's approach to establishing a DC grid bias is extremely rare in guitar amps. There is, however, a desirable side effect to this circuit: a discharge path for power supply filter capacitors that is independent of the vacuum tubes. When the amp is turned off, the voltage across the capacitors slowly bleeds away via the 220k Ω and 470 Ω resistors.

References

1Richard Kuehnel, [Guitar Amplifier Electronics: Basic Theory](#), (Seattle: Amp Books, 2018), p. 57.

2F. Langford-Smith, ed., **Radiotron Designer's Handbook**, 4th ed., (Harrison: RCA, 1953), p. 487.

3Richard Kuehnel, [Guitar Amplifier Electronics: Basic Theory](#), (Seattle: Amp Books, 2018), pp. 19-21.

4Richard Kuehnel, [Guitar Amplifier Electronics: Basic Theory](#), (Seattle: Amp Books, 2018), pp. 61-65.

5Richard Kuehnel, [Fundamentals of Guitar Amplifier System Design](#), (Seattle: Amp Books, 2019).